

Hierarchical Transformers for Motion Forecasting Based on Inverse Reinforcement Learning

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Abstract—Understanding and forecasting the future states of surrounding agents is crucial for ensuring safe and human-like decision-making for autonomous vehicles, which plays a pivotal role in the future of intelligent transportation systems. While data-driven methodologies, including recent advancements such as transformers and graph neural networks, have made significant progress, managing multimodality, achieving scene conformability, and providing human-like decision-making in future steps of the autonomy stack remains challenging. Unlike existing data-driven motion forecasting approaches that do not explicitly consider human-like motion planning, this work proposes learning from human (expert) behavior based on Inverse Reinforcement Learning (IRL). We design a novel Hierarchical Transformer-based Motion Forecasting framework, called HTMF, that learns the multimodal future motion distribution using maximum entropy IRL and leverages transformer-based learned occupancy grid maps as a context-aware approximation to the true distribution, enabling the generation of realistic motions. To assess the performance of the proposed approach, we conduct comprehensive experiments on large-scale real-world datasets across diverse challenging scenarios such as highways and unsignalized intersections. The results demonstrate that HTMF outperforms other benchmark motion forecasting methods in terms of prediction errors and in generating more plausible and human-like trajectories.

Index Terms—Autonomous driving, behavior prediction, conditional motion prediction, inverse reinforcement learning, trajectory prediction.

I. INTRODUCTION

MOTION forecasting is a pivotal component within the autonomy stack of Autonomous Vehicles (AV), playing a critical role in shaping the future of intelligent transportation. As vehicles transition towards increased automation, the ability to accurately forecast the future motions of surrounding road users, including pedestrians, cyclists, and other vehicles in their vicinity, becomes of utmost importance for ensuring safe and efficient navigation [1]. Beyond immediate navigation, motion forecasting significantly influences subsequent steps in the autonomy process, particularly in behavior planning and control [2], where providing human-like decision-making is essential for the safe

operation of AVs among humans. Consequently, the task of motion forecasting becomes crucial in enabling AVs to navigate real-world scenarios with a level of adaptability and foresight similar to human drivers, contributing to the overall safety and efficiency of autonomous driving. Early approaches to motion forecasting have merely relied on physics-based and kinematic models [3], [4] attempting to simulate the dynamics of vehicles in the environment. However, these approaches have limitations, including difficulties in accurately capturing human-like behaviors, anticipating sudden changes and adapting to diverse and evolving traffic conditions. Furthermore, since the physical characteristics of road users constantly change, these methods are more appropriate for short-term forecasting.

To alleviate these shortcomings, data-driven approaches have gained attention, providing a better understanding of human behavior, enhancing forecasting accuracy, and enabling AVs to better handle the complexities of dynamic traffic environments. However, several key challenges persist. Some data-driven approaches [5], [6] primarily focused on forecasting the motion of the target agent without explicitly considering interactions with other road users, which diverges from the reality of driving scenes where agents exhibit social behaviors. Recognizing this, interaction-aware methodologies, including models based on graph neural network [7], [8] and social pooling [9], [10] have emerged as powerful tools for capturing complex interactions within a dynamic environment. These approaches consider the influence of neighboring entities, leading to improved predictions and a more realistic representation of the driving scenario. Nevertheless, considering social interactions among various agents alone is not sufficient. Another crucial aspect involves addressing the inherent uncertainty in human behavior. To tackle this issue, most recent approaches [11], [12] leveraged the capabilities of attention mechanisms to capture diverse and complex patterns in data. These transformer-based models can handle the multimodal nature of human behavior, providing a better representation of uncertainty. Estimating a multimodal motion distribution is crucial as real-world scenarios involve multiple plausible behaviors. A multimodal approach improves risk assessment, safety, and decision-making by avoiding unrealistic unimodal/averaged predictions that may not be a feasible motion for the agent, thereby ensuring more reliable predictions of future motions. Nonetheless, while these SOTA motion forecasting approaches [5], [6], [7], [8], [9], [10], [11], [12] can improve their generalization ability by training on diverse datasets encompassing various scenarios like intersections or highways, they still tend to overlook the fundamental aspect of

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context awareness, neglecting the necessity for the estimated future motion distribution and sampled multimodal motions to be both realistic and conform to the constraints of the driving environment. This often results in predictions that fail to adhere to the road structure, leading to unrealistic outcomes such as vehicles veering off the road or into buildings and pedestrians into obstacles. Additionally, these approaches often fail to account for the complexity of the human actual motion process, leading to an inability to produce human-like motions.

To overcome these issues, we design a novel Hierarchical Transformer-based Motion Forecasting framework, namely HTMF. Our proposed approach involves approximating the distribution of agents future motions through maximum entropy Inverse Reinforcement Learning (MaxEnt IRL). Leveraging a hierarchy of transformers and a feature pyramid network, we decode Occupancy Grid Maps (OGMs) and incorporate them for a context-aware approximation to the true distribution, resulting in the generation of more realistic motions. More specifically, to achieve human-like results, we leverage insights from expert behavior, such as human driving or movements, by modeling the motion forecasting task within an IRL framework, where an agent's motion is formulated as a Markov Decision Process (MDP). To enhance the plausibility of predictions, we employ a transformer-based context encoder that combines agent dynamics with scene features. Furthermore, an attention mechanism is used to better model social interactions among agents in a driving scene and effectively connect OGMs with associated agent motion. Occupancy grids are decoded using a feature pyramid network. We condition the agent's future motions on plans sampled from the learned optimum policy, further refined by information obtained from occupancy grids. The primary contributions of this work are summarized as follows:

- We present HTMF, an innovative data-driven motion forecasting approach based on IRL that learns from expert human behavior. Our planning-based methodology approximates the true motion distribution, conditioning on plans derived from the integration of agent history and scene environment features within the Value Iteration Network (VIN)-based IRL method. This not only facilitates learning but also incorporates the human behavior process, enriching the generation of future motions.
- We enhance the accuracy of true motion approximation by predicting OGMs at every time step and incorporating them into the distribution learning process. We propose a hierarchical transformer-based method that leverages cross-attention to fuse features extracted from scene images with encoded agent history. This will improve our model capability to capture complex relationships and dependencies, allowing it to generate motions that align with the driving scene and are more feasible.
- We conduct comprehensive experiments and rigorously evaluate the performance of our proposed approach across diverse agent types and contextual environments, utilizing three large-scale motion forecasting benchmarks. The obtained results demonstrate that our model, HTMF, outperforms other baselines in terms of prediction error. This

highlights the ability of HTMF to generate human-like multimodal predictions that conform to the driving environment.

II. RELATED WORK

Predicting the future motions of diverse road users is crucial for enhancing AVs and safe interaction between humans and machines. Departing from traditional motion forecasting methodologies that primarily centered on agent kinematics [3], [13], [14], data-driven approaches broaden their perspective beyond individualistic attributes. They integrate contextual information, account for social interactions, and incorporate an understanding of uncertainty in human behavior.

As unimodal motion forecasting [15], [16] often yields outcomes infeasible for both vehicles and pedestrians, a paradigm shift towards multimodal forecasting has become imperative, reflecting the diverse motions inherent in real-world dynamics. In this direction, one approach involves directly simulating variations in the motions of road users [17], offering a comprehensive understanding of potential outcomes. Alternatively, some models concentrate on representing motion distributions through heatmaps [18], [19]. One limitation is that these approaches mainly focus on forecasting the motions of homogeneous agents. Some other models approximate motion distributions using Gaussian mixture models [5], [6], [20], [21]. For instance, Chen et al. [20] employed a Multi-Layer Perceptron (MLP) to derive embeddings for vehicle motion, which were subsequently encoded using a Long Short-Term Memory (LSTM) network. The interaction dynamics among vehicles were taken into account by the Multi-Head Attention (MHA) mechanism. The authors estimated bivariate Gaussians by employing an MLP module. This technique, however, can be prone to overfitting. Another avenue explores the implicit representation of various motion modes as latent variables [9], [10], [22], focusing on the inherent goals of road users. However, these models face limitations, including requiring multiple forward passes, challenges in real-world applicability due to randomness, concerns about interpretability, and the lack of mode diversity.

Interaction-aware motion forecasting methods have concentrated not only on modeling interactions among agents by employing social pooling-based [9], [23] and graph-based approaches [24], [25], [26] but also on capturing interactions between agents and the surrounding scene, leveraging rasterized [27], [28], [29] or vectorized feature maps [30], [31]. Tang et al. [32] utilized convolutional social (CS) pooling to model vehicle-vehicle interactions, aggregating information from neighboring vehicles to enhance forecasting. Their forecasting module comprised an LSTM-based encoder-decoder. In another work, Song et al. [33] introduced a target fusion approach in which vehicles motion histories were encoded through temporal convolutions followed by an LSTM. Spatial interactions among the vehicles were obtained using a CS pooling mechanism. However, social pooling might fall short in capturing complex spatio-temporal dependencies within dynamic scenarios. Wu et al. [7] introduced a gated fusion module that combined encoded temporal features obtained from graph attentions

with spatial feature maps computed using multi-head attention. They also generated future motions using an LSTM encoder-decoder. However, this method is not scalable and suffers from computational complexity in more significant scenes. To extract contextual information, Sadeghian et al. [34] employed a CNN over Bird's-Eye-View (BEV) images of the driving scene. The extracted features were subsequently combined with pedestrian motion observations through an attention mechanism. Lu et al. [35] suggested a dynamic map strategy that incorporated an attention mechanism to pick relevant BEV images concerning the timeframe selectively. Nevertheless, these methods might assign elevated probabilities to various unrealistic motions.

To address the shortcomings mentioned above, in our introduced model HTMF, we address multimodality by conditioning agents potential motions on plans obtained from an IRL process. Furthermore, in contrast to the approaches that rely on convolutions for contextual information extraction from BEV images and graphs for modeling interactions among agents, we benefit from transformers to perform both tasks. Furthermore, our approach emphasizes human-like motion forecasting from two key aspects: 1) the explicit use of IRL to learn directly from expert human behavior, capturing the nuanced decision-making processes of human drivers, and 2) a focus on scene conformability, ensuring that the generated motions adhere to the environment boundaries and avoid off-road areas or buildings. By incorporating these constraints, our model generates motions that are reflective of real human driving behavior. We will focus on IRL-based and transformed-based approaches in the following parts.

A. IRL-Based Motion Forecasting

The growing interest in inverse reinforcement learning (IRL) for autonomous driving, particularly in decision-making scenarios [36], stems from its promise to learn from expert demonstrations without the need for explicitly defined reward functions. In this context, IRL-based methods have been applied to two tasks. Firstly, during the planning step, IRL can assist in deriving optimal policies by discerning the latent reward structure from observed expert behaviors. Several recent works, such as those by Huang et al. [37], [38], have focused on leveraging IRL as a scoring mechanism for behavior planning. This mechanism computes the likelihood of potential behaviors based on integrated AVs and other agents behaviors. Secondly, distinct from previous methods and more related to the present study, some works [39], [40], [41], [42], [43] have employed IRL methods within the forecasting step. This can contribute to more accurate forecasting by capturing the complex patterns and tendencies inherent in human driving. In this direction, Deo et al. [39] applied maximum entropy IRL to learn an optimal policy based on context-aware rewards, subsequently sampled to generate conditional multimodal agent motions. The authors utilized BEV images of the driving scene to compute rewards using CNNs. Motions were later generated using a Gated Recurrent Unit (GRU)-based recurrent network. Our method diverges from this approach in two key aspects. While both methods utilize maximum entropy IRL, our model integrates hierarchical

transformers, which are known for their ability to capture long-range dependencies and adaptively learn hierarchical representations. Furthermore, our model use occupancy maps, which allows for a richer representation of the environment, thereby enhancing feature extraction and interaction modeling as well as improving the context-awareness of the motion predictions. Specifically, we employ two different transformers to extract comprehensive features. These embeddings are fused through cross-attention and processed through dual branches: one for predicting occupancy maps and the other focusing on learning the reward function. He et al. [42] formulated future motions of the agents as an MDP process and benefited from the value iteration network to obtain the optimal policy while integrating kinematic and contextual information from observations and driving environment. Future motion distribution was estimated using a GRU-based decoder. While these methods produce diverse multimodal motions, the majority of them are not feasible. Addressing this, Guo et al. [43] introduced an IRL-based approach, allowing the gradient computed for motion forecasting to propagate back to the behavior planning step. The authors extracted motion features using a GRU-based encoder and scene features from BEV images based on CNNs. They employed OGMs obtained using deconvolution as an intermediate step to better achieve realistic motions. Also, Yang et al. [40] employed IRL to improve scene understanding by exploring motion-scene associations. Their use of motion heat maps, which depicted agents dynamic characteristics, enhanced the alignment between motions and surrounding environment semantic segmentation.

In contrast to these approaches that predominantly address homogeneous agents, particularly pedestrians, our focus lies in enhancing motion forecasting for heterogeneous agents, encompassing vehicles, pedestrians, and cyclists. Our methodology involves utilizing transformers as a scene decoder to extract feature maps, which are subsequently employed in computing the reward function within the maximum entropy IRL framework and OGM predictions.

B. Transformer-Based Motion Forecasting

Transformers [44] have gained significant attention across various research domains for their efficiency in capturing complex spatio-temporal patterns. Their versatility in handling diverse driving scenarios and agents, combined with the interpretability provided by attention mechanisms, has sparked a growing interest in leveraging transformers for motion forecasting tasks [11], [45], [46], [47], [48], [49], [50], [51], [52], [53], [54]. In this direction, Mozaffari et al. [12] presented a maneuver-based motion forecasting approach, employing transformers to encode the motion observations of vehicles. The extracted latent representations were then utilized in both maneuver classification and motion prediction modules, generating future motions through a transformer decoder. However, this method suffers from the limitation of not considering scene conformability. Feng et al. [47] utilized a transformer-based context encoder, employing self-attention to extract interactions within vectorized scene information and agent history, while cross-attention was used to fuse information between the two

sources. Nasr Azadani et al. [41] benefited from transformers in their forecasting module, where future motions were conditioned on context information obtained from an IRL process. Sampled plans were then fed to a transformer-based encoder-decoder for target and history fusion and motion refinement. Liu et al. [48] designed a transformer-based architecture that aggregated vehicles motion history and scene encodings over static motion proposal. Chen et al. [53] took advantage of transformers for social reasoning within a conditional variational autoencoder framework. They defined two-stream encoder modules that create graph transformers to extract social and temporal encodings. In another work, Chen et al. [54] presented a goal-based motion forecasting approach in which they combined the representation learning abilities of transformers and GRU-based recurrent networks to capture the temporal relationships across various time steps. Vishnu et al. [49] introduced a conditional variational autoencoder that employs transformers at its core for both encoding and decoding. To include temporal information for each vehicle, they replaced the position encoder in transformers with a time encoder. However, this method inherits limitations from approaches using latent variables as priors, such as the need for multiple forward passes and a lack of mode diversity. Gao et al. [11] proposed a joint intention and motion prediction approach, with each utilizing transformers for latent reasoning and modeling social interactions. However, this method overlooks the fundamental aspect of context awareness, neglecting the necessity for the estimated future motion distribution and sampled multimodal motions to be both realistic and in conformity with the driving environment and human-like motions.

In contrast, different from these approaches, we employ visual transformers for scene latent encoding extractions, transformers for motion history, and socially aware cross-attention for information fusion, which are further used for OGMs and, ultimately, conditional motion forecasting.

III. PRELIMINARIES

In this section, we provide a concise overview of Markov Decision Processes and MaxEnt IRL applied to motion forecasting.

MDP. In motion forecasting, modeling the dynamics of an environment where entities move is crucial. Leveraging the MDP framework allows us to encapsulate this complexity. Here's a brief overview of key MDP components:

- States ($\mathcal{S}$) might represent distinct situations an agent may find itself in, such as the positions of agents on the map or grids defined over the driving scene.
- Actions ($\mathcal{A}$) constitute the finite set of choices available to the agent in each state, denoting the set of permissible movements or decisions an agent can make, e.g., moving left or right.
- State transition matrix ($\mathcal{T} : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$) denotes the likelihood of transitioning from one state (s) to another (s') by taking a specific action (a), reflecting the dynamic nature of motion forecasting.
- Reward function ($\mathcal{R}$) assigns a numeric value, capturing the immediate cost associated with taking an action. For

instance, it could determine where to navigate or how to avoid obstacles, contributing to the creation of human-like motions in the context of motion forecasting.

MaxEnt IRL. Reinforcement learning tasks are modeled as MDPs, providing a structured framework to describe the interaction between an agent and its environment. In the context of inferring an unknown reward function, MaxEnt IRL emerges as a powerful approach [55], combining principles from MDPs with the concept of maximum entropy. In an MDP, a motion τ can be represented as a set of state-action pairs. Let us denote the motion as $\tau = \{(s_0, a_0), (s_1, a_1), \dots, (s_T, a_T)\}$, where s_t represents the state at time step t and a_t represents the action taken at time step t . Each pair (s_t, a_t) corresponds to a state-action pair at a specific time step in the motion. The representation of the reward $\mathcal{R}$ for a motion τ within the MaxEnt IRL framework is characterized by the summation of rewards at each time step throughout the motion. This formulation can be expressed as:

$$\mathcal{R}(\tau) = \sum_t r(s_t) = \alpha^T f(\tau) \quad (1)$$

where $r(s_t)$ signifies the reward at time step t , $f(\tau)$ is the feature vector along the motion τ , and α denotes the reward weights. Note that no discount factor is considered since the core principle here is to maximize the entropy of the policy given the constraints imposed by the observed behavior. This principle encourages exploration and diversity in inferred policies, ensuring that the model does not overly commit to specific motions unless strongly supported by the data. Introducing a discount factor could bias the inferred policy towards certain actions or states, reducing the entropy and potentially leading to less diverse and less accurate modeling of the expert behavior. Moreover, the experts behavior is observed over finite time horizons. In such settings, the experts motions are finite and do not require discounting future rewards to keep the value function bounded. Furthermore, MaxEnt IRL assumes a stochastic policy that models human motion distribution by a Boltzmann distribution in which the probability of a motion given the reward weights can be formulated as:

$$P(\tau|\alpha) = \frac{e^{\alpha^T f(\tau)}}{Z(\alpha)} \quad (2)$$

where $Z(\alpha)$ is a normalizing factor. Since $Z(\alpha)$ is often challenging to compute due to the need for integration over the entire motion space, a practical approximation is employed. By generating a finite set of feasible motions, Z is approximated, resulting in a more manageable probability mass function:

$$P(\tau|\alpha) \approx \frac{e^{\alpha^T f(\tau)}}{\sum_{i=1}^N e^{\alpha^T \tilde{f}_i}} \quad (3)$$

where $\tilde{f}_i$ represents the feature vector of a generated motion, and N is the number of produced motion.

The primary objective of MaxEnt IRL is to optimize the reward weights α to maximize the likelihood of human behaviors $\mathcal{D} = \{\tau_1, \tau_2, \dots, \tau_N\}$ within the motion distribution, where $\tau = \{(s_0, a_0), \dots, (s_T, a_T)\}$. The cost function $J(\alpha)$ is defined

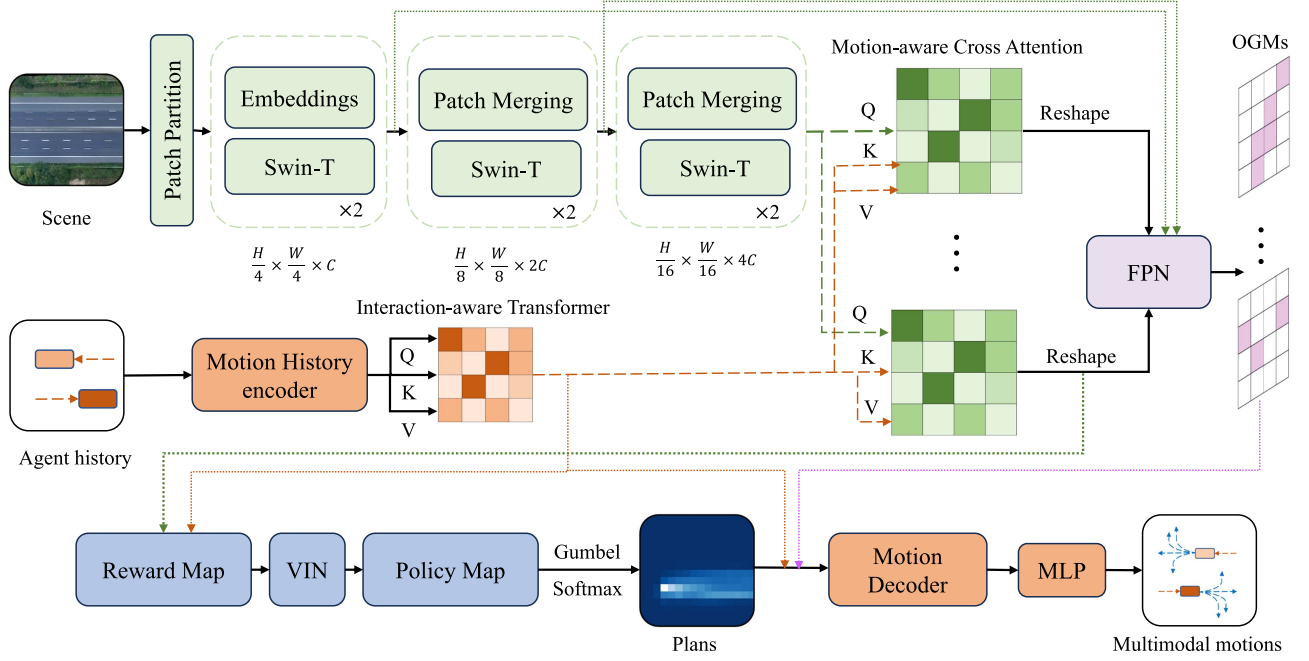


Fig. 1. An overview of HTMF: (1) Swin-Transformer and history encoder embed scene and expert information, (2) A feature pyramid decoder predicts OGMs based on features extracted by integrating scene and history through cross attention, (3) Policy maps are derived by employing VIN approximation, and (4) Transformers and MLP are utilized to decode future motions.

as the log-likelihood of the expert behavior set:

$$J(\alpha) = \max_{\alpha} \sum_{\tau \in \mathcal{D}} \left(\alpha^T f(\tau) - \log \sum_{i=1}^N e^{\alpha^T \tilde{f}_i} \right) \quad (4)$$

Note that α represents the parameters of the reward function and is a continuous variable, i.e., it lies in a continuous parameter space. This characteristic allows the values of α to take any real number, enabling smooth adjustments during optimization. As a result, this cost function can be optimized using a gradient-based method:

$$\nabla J(\alpha) = \sum_{\tau \in \mathcal{D}} f(\tau) - \sum_{i=1}^N \frac{e^{\alpha^T \tilde{f}_i} \tilde{f}_i}{\sum_{j=1}^N e^{\alpha^T \tilde{f}_j}} \quad (5)$$

IV. METHODOLOGY

This section provides an overview of the problem formulation for motion forecasting and introduces our proposed architecture, namely HTMF.

A. Problem Formulation

Forecasting the future motions of heterogeneous agents within a driving environment can be formulated within the framework of conditional probability, where the goal is to estimate the distribution of forthcoming motions, considering the past and present states of all agents, along with contextual information from the scene. Specifically, in a scenario involving N agents, each agent i is characterized by a motion history $X_i = \{(\eta_{i,0}, \eta_{i,1}, \dots, \eta_{i,T})\}$, and ζ represents contextual information extracted from the scene. The task involves forecasting the future

motions of all agents within a time horizon ΔT , denoted as $\hat{Y}_i = \{(\eta_{i,T+1}, \eta_{i,T+2}, \dots, \eta_{i,T+\Delta T})\}$. The conditional probability distribution for these future motions is expressed as:

$$P(\hat{Y}_i | X_i, \zeta), \quad (6)$$

where the probability distribution is conditioned on the historical motion of the agent X_i and the contextual information ζ derived from the scene. We formulate the forecasting task as an MDP, wherein the future motions of agents are contingent on grid-based states gained from optimal human behavior. In our formulation, agents motion are considered as state-action pair $\eta_i = \{(s_{i,0}, a_{i,0}), (s_{i,1}, a_{i,1}), \dots, (s_{i,T}, a_{i,T})\}$ and agents employ a non-stationary policy to dynamically adapt and assess optimal actions at each MDP step, effectively predicting future behavior by estimating state sequences. This mathematical model is succinctly represented as a set $(\mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R})$, encompassing possible states $\mathcal{S}$ as cells on a defined grid, actions $\mathcal{A}$ comprising movements, including going towards main directions on the grid, a matrix for state transitions $\mathcal{T}$, and reward functions $\mathcal{R}$ for end targets and the paths.

B. Proposed Architecture

1) *Overall Framework*: An overview of our proposed model architecture is illustrated in Fig. 1. As demonstrated generally, by utilizing a combination of contextual information and agent dynamics, we learn the MDP reward function and predict occupancy maps. The obtained rewards are further utilized to determine the optimal policy within the proposed MaxEnt IRL framework. Predicted occupancy maps and plans sampled from the optimal policy are further utilized to estimate the motion

distributions. More specifically, in our introduced model, we take advantage of hierarchical transformers to extract scene and history feature embeddings along with capturing their interactions. We employ the Swin-transformer to extract relevant features from BEV images of the surrounding environment. Additionally, kinematic encodings are obtained using attention mechanisms. These extracted embeddings are then combined through cross-attention, reshaped, and fed into two branches. The first branch utilizes a Feature Pyramid Network to forecast occupancy maps, serving as auxiliary information. Simultaneously, the second branch is responsible for learning the reward function. To learn the optimal policy, we employ value iteration approximation, and to extract the potential plans, we utilize the Gumbel Softmax sampling method. Subsequently, multimodal future motions and their respective probabilities are generated using transformers and a multi-layer perceptron.

2) *Scene Representation*: A traffic scenario involves both agents and scene information. Examining the scene information can significantly improve situational awareness, leading to enhanced scene adaptability and conformity. In order to systematically represent this scenario, we extract feature entities that include the motion representation of road users and feature maps from the image of the scene, denoted as I . These entities correspond to both temporal and spatial aspects of scene representation. Utilizing an agent-centric reference frame, we employ the relative positions and velocities of the agents, denoted as (x, y, v_x, v_y) .

3) *Encoder*: The primary module of HTMF incorporates a context encoder, which consists of an agent history encoder for extracting feature maps from the expert behavior of the target and its surrounding agents. It also includes a visual encoder to derive environment embeddings from BEV images of the driving scene.

Visual Encoder: The Visual encoder module is responsible for extracting the scene encodings from the scene image I . To achieve this, we employ a vision transformer [56] due to its flexibility in learning representations at different scales and its linear computational cost w.r.t. the dimension of I . In the initial step, the scene image is divided into non-overlapping patches and embedded into a shape of $H/4 \times W/4 \times C$. Subsequently, as shown in Fig. 1, three blocks of Swin Transformer (Swin-T) blocks are applied over the obtained patches. The detailed composition of each block is depicted in Fig. 2(a). As demonstrated, each Swin-T block comprises hierarchical transformers employing window and shifted window-based Multi-head Self-Attention (MSA) denoted as ϕ_w and ϕ_{sw} , followed by layer normalization (LN) and an MLP. These blocks play a crucial role in learning interaction representations while extracting scene features. The computations of these blocks are expressed as:

$$\hat{g}^l = \phi_w(LN(g^{l-1})) + g^{l-1} \quad (7)$$

$$g^l = MLP(LN(\hat{g}^l)) + \hat{g}^l \quad (8)$$

$$\hat{g}^{l+1} = \phi_{sw}(LN(g^l)) + g^l \quad (9)$$

$$g^{l+1} = MLP(LN(\hat{g}^{l+1})) + \hat{g}^{l+1} \quad (10)$$

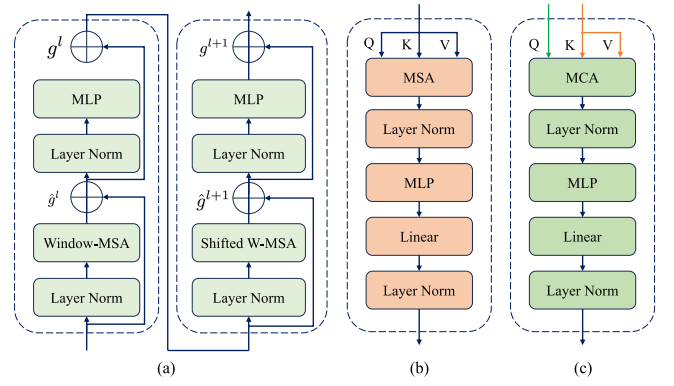


Fig. 2. Detailed architecture of (a) swin-T block, (b) motion transformer, and (c) cross attention fusion module.

The output of the visual encoder $\Omega_s \in \mathbb{R}^{H/16 \times H/16 \times 4C}$ is the scene encodings in different scales:

$$\Omega_s = \text{Swin-T}(I) \quad (11)$$

History Encoder: In parallel with the visual encoder, we simultaneously capture representations of agent motion histories and their interactions using a transformer module. Specifically, in a scenario with N agents, we encode the motions of agents, denoted as $\eta \in \mathbb{R}^{N \times (T+1) \times 4}$, through a Multihead Self-Attention (MSA) module, followed by a global pooling layer. The input is the agent speed, along with the longitudinal and lateral location information corresponding to grid cells. The output of this process is denoted as $\beta \in \mathbb{R}^{N \times 4C}$. Subsequently, we analyze their interactions utilizing the transformer structure illustrated in Fig. 2(b). All layers maintain consistent dimensions for latent representations. For each attention head h , and given the projected query Q , key K , and value V vectors, the attention score a_h and weights $W_{\text{att},h}$ are computed as follows:

$$\text{Head}_i = \text{softmax} \left(\frac{Q_h K_h^T}{\sqrt{d_k}} + b_{\text{att},h} \right) V_h$$

$$\text{MSA}(Q, K, V) = (\text{head}_1 || \dots || \text{head}_H) W \quad (12)$$

where d_k is the dimension of the key vectors, and $b_{\text{att},h}$ is the bias for the attention weights for head h . Having the output of MSA followed by a global pooling layer β , the final output of the motion history encoder $\Omega_m \in \mathbb{R}^{N \times 4C}$ is calculated as:

$$\Omega_m = \text{Transformer}(\beta[x, y, v_x, v_y]) \quad (13)$$

Cross Attention: After obtaining latent representations for both the scene Ω_s and motion history Ω_m , we employ a Multi-head Cross Attention (MCA) mechanism to integrate information across these two components. MCA enables the model to link each grid cell with kinematic details derived from the agents, enhancing the flow of information. As demonstrated in Fig. 2(c), we employ scene-extracted representations as queries and extracted motion entities as keys and values. The output of this module $\Omega_c \in \mathbb{R}^{H/16 \times H/16 \times 4C}$ is computed as:

$$\Omega_c = \text{MCA}(\Omega_s, \Omega_m) \quad (14)$$

4) *OGM Decoder*: To explicitly approximate the distribution of true expert behavior, we utilize a Feature Pyramid (FP) network [57] to anticipate a series of dynamic OGMs. The OGM decoder is responsible for reconstructing or decoding occupancy information O_t at time t from the fused features Ω_c obtained in the previous stage. To decode the occupancy maps, we take advantage of a pyramid of 2-D CNNs. The same set of 2-D CNNs is used across multiple future time steps, i.e., the weights of the convolutional layers are shared, promoting parameter sharing and allowing the network to learn common features across different instances in time. Another set of 2-D CNNs is used to analyze features within the residual connections. This enables the network to focus on capturing and processing the residual information independently, learning and representing unique features that deviate from the shared features. Our OGM decoder module can model the transition process of the likelihood distribution of the agents characteristics. Predicted OGMs are further employed in our multimodal decoder module as auxiliary information to enhance the plausibility of the motions.

5) *Policy Network*: After combining the motion and scene latent representations, we adopt MaxEnt IRL to solve the MDP problem. We transform these representations into IRL motion and end target rewards $(\mathcal{R}_\alpha^m, \mathcal{R}_\alpha^t)$ employing CNNs. The motion rewards assist in selecting an action, while the target rewards are employed to signify the final point of motion. Consequently, as actions with higher rewards are linked to increased probability, the optimal motion is learned by maximizing the log-likelihood of expert behavior, denoted as $\mathcal{H} = \cup_{k=1}^K \eta_k$. This learning process, as demonstrated by Zhang et al. [15], is differentiable w.r.t. α . The entire process can be succinctly expressed as:

$$\begin{aligned} \arg\max_{\alpha} J(\alpha) &= \arg\max_{\alpha} \sum_{\eta_k \in \mathcal{H}} \log P(\eta_k | \mathcal{R}_\alpha) \\ P(\eta_k | \mathcal{R}_\alpha) &= \frac{e^{\mathcal{R}_\alpha(\eta)}}{Z(\alpha)} \\ \nabla J(\alpha) &= \sum_{\eta \in \mathcal{H}} (\mu - \hat{\mu}) \nabla R(\alpha) \end{aligned} \quad (15)$$

where α represents the reward parameters, μ and $\hat{\mu}$ represent the true and expected likelihood of visiting different states in a given environment during the execution of a particular behavior η .

The policy network module of HTMF is responsible for identifying the optimal policy π_α by utilizing computed rewards and sampling this policy to generate plans. These plans condition the estimation of the motion distribution. We apply a VIN approximation, as detailed in Algorithm 1, to derive a policy given the obtained motion and end target rewards $(\mathcal{R}_\alpha^m, \mathcal{R}_\alpha^t)$. The resulting policy $\pi_{\alpha,n}(a|s)$ represents the probability of performing a movement in a given state, accommodating time-dependent MDP time steps. As described in Algorithm 1, the update process involves recursive adjustments to the log partition values $Q^i(s, a)$ and V_s^i . This updating mechanism serves as an approximation for non-stationary rewards. At each step, the policy is determined through a calculation where the probability $\pi_{\alpha,n}(a|s)$ of taking a in a given s is determined by the

Algorithm 1: Approximate Value Iteration.

Input: $\mathcal{R}_\alpha^m, \mathcal{R}_\alpha^t$

Output: $\pi_{\alpha,n}(a|s)$

```

1  $V_s \leftarrow -\infty, \forall s \in \mathcal{S}_m$ ;
2 for  $n \in \{N, \dots, 2, 1\}$  do
3    $V_{s,n} \leftarrow \mathcal{R}_\alpha^t, \forall s \in \mathcal{S}_t$ ;
4    $Q_n(s, a) = \mathcal{R}_\alpha^m + V_{s',n} = \mathcal{T}(s, a) \forall a \in \mathcal{A}, \forall s \in \mathcal{S}_m$ ;
5    $V_{s,n-1} = \log \sum_{a \in \mathcal{A}} e^{(Q_n(s,a))}, \forall s \in \mathcal{S}_m$ ;
6    $\pi_{\alpha,n}(a|s) = e^{(Q_n(s,a) - V_{s,n})}$ ;
end
return:  $\pi_{\alpha,n}(a|s)$ 

```

Algorithm 2: Policy Propagation.

Input: π_α

Output: $\hat{\mu}$

```

1  $\hat{\mu}_s^1 \leftarrow 0, \forall s \in \mathcal{S}$ ;
2  $\hat{\mu}_{s_b}^1 \leftarrow 1, s_b = \text{initial state}$ ;
3 for  $n \in \{1, 2, \dots, N\}$  do
4    $\hat{\mu}_{s,n+1} = \sum_{a,s'} \pi_{\alpha,n}(a|s') \hat{\mu}_{s',n}, \mathcal{T}(s', a) = s$ ;
end
5  $\hat{\mu}_s = \sum_n \hat{\mu}_{s,n}$ ;
return:  $\hat{\mu}_s$ 

```

exponential of the difference between Q_n and $V_{s,n}$. Afterward, we use Algorithm 2 to calculate the expected likelihood of state visitation $\hat{\mu}$. Policy propagation involves iteratively applying the optimal policy to produce $\hat{\mu}$ values, starting from the initial state s_b . Finally, we take advantage of a Gumbel Softmax to sample the optimal policy due to its stability capability while training the model. This process is formulated as:

$$\mathcal{G}_d = \frac{\exp\left(\frac{\log(z_d) + c_d}{\lambda}\right)}{\sum_{l \in d} \exp\left(\frac{\log(z_l) + c_l}{\lambda}\right)}, \quad \text{for } d = 1, \dots, D \quad (16)$$

where z_d denotes the d -th element of the input vector representing the logits, c_d represents an i.i.d sample, and λ serves as a factor influencing the level of sharpness in the distribution. The sampled plans are anticipated to encompass diverse end targets and paths leading to these targets, thereby facilitating the generation of scene-compliant motions.

6) *Multimodal Decoder*: The Multimodal decoder is tasked with generating the potential motion distribution, given a sample p_i from the optimal policy along with scene and motion information. We employ transformers for our motion decoder, where the input to this module comprises predicted OGMs O_t , motion encodings Ω_m , and samples from the policy network p_i . The output consists of the future multimodal motions of the agents, encompassing K modes. Initially, a transformer is employed to integrate contextual information from p_i with motion features of

surrounding agents, including speed and acceleration, into their respective grid cells and state locations. The agents previous states serve as the query Q_i , while plan features are considered as the key K_i and values V_i in the attention mechanism. Subsequently, an MCA mechanism is utilized to concatenate the hidden maps of O_t with corresponding context feature maps. In the final step, we generate multimodal outputs using an MLP structure, resulting in $\hat{Y}_i = \{(\eta_{i,T+1}, \eta_{i,T+2}, \dots, \eta_{i,T+\Delta T})\}$, where ΔT represents the prediction window, conditioned on the sampled plans.

7) *Loss Function*: In our learning framework, the OGM learning is enhanced by adopting focal loss ($\mathcal{L}_F$) [58] as the primary loss function. This is motivated by the desire to effectively address class imbalance in our OGM decoder:

$$\mathcal{L}_F(p, O) = -\frac{1}{N} \sum_{i=1}^N [(1 - p_i)^\gamma \cdot \log(p_i)] \quad (17)$$

where γ is the focusing parameter. Moreover, as for the motion distribution, we use symmetric focal loss to guide the policy network and motion decoder training. This loss function encompasses Focal Loss applied to both the predicted and true motion distributions. Finally, we propose training the model leveraging the variety loss. Minimizing this loss encourages the model to generate representative motions that capture different plausible outcomes.

$$\mathcal{L}_v = \min_{k \in \{1, \dots, K\}} \|Y - \hat{Y}^{(k)}\|_2 \quad (18)$$

where K represents the number of sampled motions.

V. EXPERIMENTAL EVALUATION

In our experimental evaluation section, we will discuss the datasets we used, the metrics employed for evaluation, the implementation details of our model, and the results obtained from our experiments. This comprehensive overview will provide insights into the performance and effectiveness of HTMF across different scenarios.

A. Dataset

We examine the performance of our proposed approach HTMF over three diverse scenarios and agent types. This will allow us to thoroughly compare HTMF performance with state-of-the-art methods of motion forecasting. Each of these datasets is briefly discussed in the following.

- *HighD dataset*: This dataset [59] consists of data on German highways, an example of structured scenarios. Collected using drones across six distinct locations, the dataset spans more than 45,000km, providing a rich source of naturalistic driving data. This dataset comprises over 110 thousand trajectories involving a variety of agents, including cars and trucks. The recorded information includes detailed data on locations, speed, and acceleration, as well as the type and size of vehicles. The BEV images of the roads are also available. Following previous models [11], [60], we reduce the sampling rate by a factor of 5. Furthermore, the

observation window is set at 3 seconds, while the prediction window extends up to 5 seconds.

- *InD dataset*: This dataset [61] encompasses trajectories from five distinct agent types, including both vehicles and vulnerable road users, gathered from unsignalized intersections in Germany. This dataset exemplifies unstructured and complex driving scenarios, which inherently pose greater challenges in accurately forecasting the motions of diverse agents. Collected from four different intersections, the dataset comprises over 13,000 trajectories recorded at a frequency of 25 Hz. The provided information includes BEV images of the scenes and dynamic details of the agents involved. Following previous studies [21], [24], [26], [41], we reduce the sampling rate by a factor of 10 for a fair comparison. Moreover, we adopt an observation window of 3.2 seconds (8 frames) and a prediction window of 4.8 seconds (12 frames) unless specified otherwise.
- *SDD dataset*: This dataset [62] consists of diverse data from heterogeneous agents, encompassing both vehicles and vulnerable road users. The dataset was captured using drones at Stanford University, comprising 60 BEV videos recorded at a frequency of 2.5 Hz. Notably, this dataset serves as an exemplar of scenarios marked by high interactions between agents, featuring an extensive collection of over 185,000 interactions. The inclusion of varied scenes within the dataset provides a valuable means to assess the generalization ability of models. The evaluation setting for this dataset is consistent with that of prior studies [39], [41], [52] for fair comparison. Furthermore, the observation and prediction windows are set at 12 and 8 frames, equivalent to 4.8 and 3.2 seconds, respectively.

B. Metric

In accordance with prior research on the utilized datasets, we employ the following error metrics commonly adopted to evaluate the effectiveness of motion forecasting methods.

- *Root Mean Square Error (RMSE)*: To facilitate comparisons with state-of-the-art methods, we employ the RMSE metric for evaluations carried out over the HighD dataset. The RMSE is calculated based on the prediction that has the highest probability and is computed as follows:

$$RMSE = \sqrt{\frac{1}{\Delta T} \sum_{t=T+1}^{T+\Delta T} (\hat{Y}^t - Y^t)^2} \quad (19)$$

- *Average Displacement Error (ADE)*: It measures the average Euclidean distance between the predicted motions ($\hat{Y}_k^t$) and the corresponding true motions (Y^t) for each target agent over a specified prediction window ΔT . We calculate ADE as follows:

$$ADE = \frac{1}{K \cdot \Delta T} \sum_{k=1}^K \sum_{t=T+1}^{T+\Delta T} \|\hat{Y}_k^t - Y^t\|_2 \quad (20)$$

- *Final Displacement Error (FDE)*: This metric evaluates the accuracy of predictions at the final time step within a ΔT . It measures the Euclidean distance between the forecast final

TABLE I
RMSE RESULTS OF STRUCTURED HIGHWAYS SCENARIO OVER HIGHD DATASET FOR SEVERAL BENCHMARK METHODS

Horizon (s)	Social-LSTM [10]	CS-LSTM [23]	Social-GAN [9]	PiP [33]	Tang et al. [32]	STDAN [20]	MHA [63]	TP2NET [60]	Gao et al. [11]	MMnTP [12]	HTMF
1	0.22	0.22	0.30	0.17	0.20	0.19	0.06	0.05	0.41	0.19	0.06
2	0.62	0.61	0.78	0.52	0.59	0.27	0.09	0.07	0.79	0.38	0.08
3	1.27	1.24	1.46	1.05	1.21	0.48	0.24	0.19	1.11	0.62	0.14
4	2.15	2.10	2.34	1.76	2.08	0.91	0.59	0.49	1.40	0.95	0.43
5	3.41	3.27	3.41	2.63	3.24	1.66	1.18	0.98	-	1.39	0.78

The observation window is 3s. Best results are shown in bold.

motions ($\hat{Y}_k^{T+\Delta T}$) and the actual final motion ($Y^{T+\Delta T}$) for each agent. We compute FDE as:

$$FDE = \frac{1}{K} \sum_{k=1}^K \|\hat{Y}_k^{T+\Delta T} - Y^{T+\Delta T}\|_2 \quad (21)$$

- *Off-road Rate*: This metric evaluates how accurately the predicted motions align with the actual road structures or permissible areas. It indicates the number of non-scene-compliant predictions.

In all four metrics, a smaller value indicates higher performance.

C. Setting

We train our model, HTMF, using the Adam optimizer with an initial learning rate of 10^{-4} and a decay rate of 50% applied every 25 epochs, for a total of 100 epochs. The batch size is set to 32, and the model is trained on two NVIDIA H100 GPUs. The embedding size of the MLP is 64. For activation functions within the encoder and FPN module, GELU and ELU are employed, respectively. To address overfitting, a dropout mechanism is incorporated after every MLP layer, each set with a rate of 0.1. The BEV images are cropped around the agents with a size of 200, and data augmentation is applied to create agent-centric frames. We use 8 attention heads for both MSA and MCA mechanisms. From the optimal policy map, 500 plans are sampled, and we generate 20 multimodal predictions as output.

D. Baseline Methods

We compare the performance of HTMF with several state-of-the-art approaches, each of which is briefly discussed in the following.

- *Structured scenarios*: For this scenario, we conduct a comprehensive comparison with state-of-the-art benchmarks on the HighD dataset, as detailed in Section V-A. We choose 10 recent models that employ three distinct approaches to capture social interactions among agents. First, we compare the results of HTMF with those of five social pooling-based techniques, including Social-LSTM [10], CS-LSTM [23], Social-GAN [9], PiP [33], and Tang et al. [32]. Second, we compare HTMF performance with that of attention-based methodologies, which benefit from attention mechanism in an LSTM-based encoder-decoder architecture, including MHA [63] and TP2NET [60].

Moreover, our comparison consists of recent transformer-based approaches such as Gao et al. [11], STDAN [20], and MMnTP [12].

- *Dynamic scenarios*: In this scenario, we compare the performance of HTMF with several state-of-the-art models over unstructured intersections of the Ind dataset and dynamic scenes of the SDD dataset. Regarding the former, we compare HTMF efficiency with that of two recent IRL-based and transformer-based approaches, Guo et al. [17] and CAPHA [41]. In addition, we take advantage of two pooling-based techniques, Social-LSTM [10] and Social-GAN [9], Grip++ [25] which models social interactions among various agents within a scene using graphs, and finally five attention-based approaches, including DCENET [64], SCOUT [24], STAG [26], MTP-GO [21], and StarNet [22]. Considering the latter, we use six benchmarks including two transformer-based approaches (MSN [52] and MID [17]), two transformer and IRL-based models (CAPHA [41] and Guo et al. [43]), one IRL-based technique namely P2T [39], and finally one goal-based approach called Y-Net [28].

E. Quantitative Evaluation

Highways: We assess the effectiveness of our model, HTMF, on the HighD dataset and perform a comparative analysis against several benchmark models discussed in Section V-D. Table I provides a summary of the prediction errors for HTMF and 10 other approaches, measured in terms of RMSE, where the historical horizon is 3 seconds, and the forecasting horizon ranges from 1 to 5 seconds. All the compared approaches incorporate social reasoning. Notably, pooling-based approaches exhibit lower performance than other models employing social reasoning. In this category, Tang et al. [32] enhance baseline pooling-based approaches, such as Social-LSTM, through the utilization of convolutional social pooling. Additionally, the MHA and TP2NET models improve results by leveraging attention mechanisms, where TP2NET achieves the best performance in short-term predictions with a 0.05 and 0.07 RMSE for 1 s and 2 s forecasting horizons, respectively. Compared to previous methodologies, Gao et al. [11] and MMnTP, utilizing transformers, demonstrate superior results by effectively capturing intricate social interactions and dependencies among agents. This enables more accurate predictions and a heightened understanding of structured scenarios. As demonstrated, HTMF outperforms all other approaches in long-term predictions by

TABLE II
ADE/FDE OVER IND SCENARIOS WITH OBSERVATION AND PREDICTION WINDOW OF 3.2 s AND 4.8 s

Scenario	Social-LSTM [10]	Social-GAN [9]	Grip++ [25]	DCENet [64]	SCOUT [24]	STAG [26]	CAPHA [41]	HTMF
Bendplatz	1.28 / 3.19	1.55 / 3.23	0.94 / 2.06	0.64 / 1.41	0.48 / 1.08	0.37 / 0.66	0.25 / 0.54	0.23 / 0.46
Frankenburg	1.78 / 4.24	2.22 / 4.45	0.59 / 1.41	0.86 / 1.93	0.30 / 0.69	0.49 / 1.06	0.17 / 0.48	0.15 / 0.36
Heckstrasse	2.17 / 5.11	2.71 / 5.64	1.94 / 4.46	0.28 / 0.62	0.40 / 0.83	0.30 / 0.48	0.19 / 0.39	0.19 / 0.31
Strasse	2.29 / 5.33	3.02 / 5.30	1.65 / 3.65	0.96 / 2.12	0.67 / 1.55	0.51 / 1.30	0.56 / 1.11	0.48 / 0.99
Average	1.88 / 4.47	2.38 / 4.66	1.28 / 2.88	0.69 / 1.52	0.46 / 1.03	0.42 / 0.88	0.29 / 0.63	0.26 / 0.53

TABLE III
EVALUATION RESULTS (ADE / FDE) ON VARIOUS AGENT TYPES OVER IND DATASET

Agent Type	Guo et al. [43]	MTP-GO [21]	SCOUT [24]	STAG [26]	StarNet [22]	CAPHA [41]	HTMF
Vehicles	-	1.07 / 3.32	0.36 / 0.80	0.35 / 0.69	0.32 / 0.89	0.28 / 0.61	0.25 / 0.54
Pedestrians	1.97 / 2.98	0.38 / 1.03	0.52 / 1.16	0.38 / 0.90	-	0.27 / 0.82	0.21 / 0.73
Cyclists	-	0.90 / 2.65	0.47 / 1.06	1.28 / 2.84	-	1.25 / 2.65	0.85 / 2.03

TABLE IV
ADE / FDE RESULTS ON VARIOUS AGENT TYPES OVER IND SCENARIOS

Scenario	Vehicles	Pedestrians	Cyclists	All
Bendplatz	0.28 / 0.52	0.31 / 0.52	0.12 / 0.28	0.23 / 0.46
Frankenburg	0.08 / 0.24	0.12 / 0.28	0.54 / 1.32	0.15 / 0.36
Heckstrasse	0.10 / 0.22	0.09 / 0.28	0.44 / 1.21	0.19 / 0.31
Strasse	0.48 / 1.15	0.41 / 0.68	1.58 / 4.02	0.48 / 0.99

TABLE V
RESULTS COMPARISON OVER SDD DATASET

Models	ADE	FDE	off-road rate
P2T [39]	8.78	14.08	0.060
Y-net [28]	7.85	11.85	0.048
MSN [52]	7.68	12.16	-
CAPHA [41]	7.67	11.41	0.042
MID [17]	7.61	14.30	-
Guo et al. [43]	6.77	10.46	0.066
HTMF	6.48	10.52	0.031

combining IRL and transformer strength. Conditioning the future motions of agents on plans sampled from optimal policy OGMs yields remarkable results. For instance, when compared to the second-best approach and considering a 5 s prediction horizon, HTMF exhibits an improvement of 20%, achieving an RMSE of 0.78.

Unsignalized intersections: In this scenario, we conduct a comparative analysis of HTMF effectiveness against recent benchmarks on the InD dataset, utilizing ADE/FDE as evaluation metrics. As depicted in Table II, our approach consistently outperforms all comparison models across all four unsignalized intersections. Specifically, at Strasse intersections, HTMF attains an ADE of 0.48 and an FDE of 0.99, showcasing a substantial improvement over the second-best model by 14% and 11%, respectively. Notably, DCENet, which

TABLE VI
ABLATION STUDY RESULTS OVER SDD DATASET

Reward maps	Policy maps	Motion decoder	ADE	FDE	Off-road rate
×	×	✓	8.82	14.26	0.055
✓	✓	×	7.25	11.51	0.037
×	✓	✓	6.57	10.57	0.038
✓	✓	✓	6.48	10.52	0.031

conditions multiple motion modes on latent variables, exhibits superior error values compared to pooling-based models such as Social-GAN. This highlights the importance of addressing multimodality while forecasting the potential motions of heterogeneous agents. SCOUT enhances performance using a graph attention-based social reasoning approach, achieving an average of 0.46/1.03 in terms of ADE/FDE. STAG also takes advantage of graph attention. However, it enhances the error values by constructing sparse graphs for various agents. CAPHA improves their results by generating scene-compliant predictions. HTMF achieves state-of-the-art results by leveraging both IRL and transformer structures, with average errors of 0.26/0.53 in terms of ADE/FDE.

In this scenario, we extend our performance comparison to various types of agents across all intersections in the InD dataset, evaluating several benchmark models. This allows us to analyze the influence of different agent types on the performance of forecasting models, as different agent groups can represent unique patterns. As demonstrated in Table III, HTMF consistently outperforms all other benchmarks in forecasting the future motions of both pedestrian and vehicle agent types. Take vehicles as an example; compared to attention-based approaches, our results show a notable improvement of 22% in both ADE and FDE error values. For pedestrians, HTMF also demonstrates enhanced performance compared to the most recent IRL-based and transformer-based approaches, such as

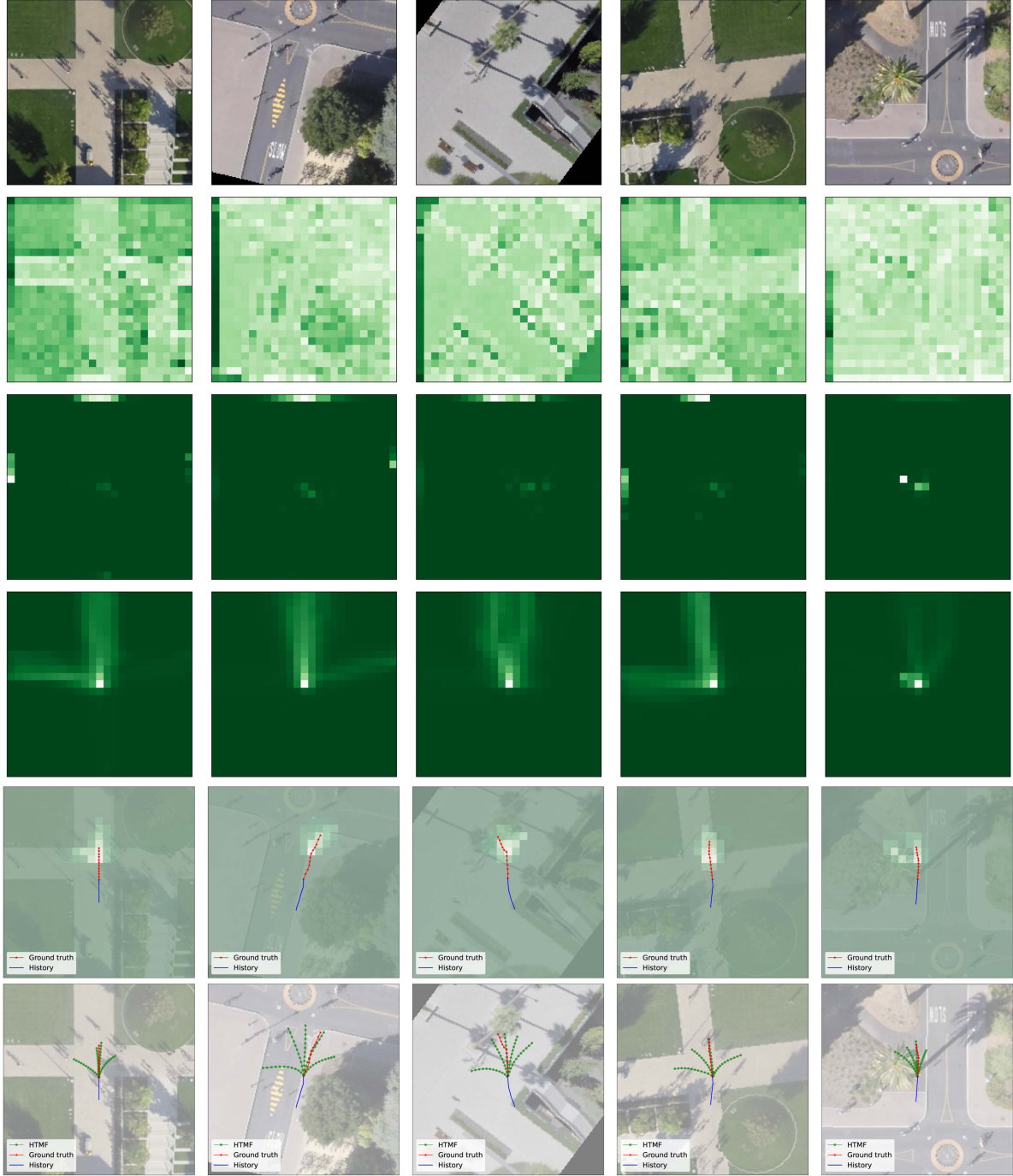


Fig. 3. Visualized predictions of our proposed model, HTMF, over various SDD scenes. From top to bottom: (1) the BEV image of the scene, (2) the reward map, (3) the ultimate goals, (4) the paths to the goals, (5) OGM predictions at $t = 8$, and (6) the multimodal predicted motions with $K = 5$.

Guo et al. [43] and CAPHA. This highlights the effectiveness of leveraging transformer-based encoders within visual and history encoders. HTMF achieves a significant improvement of 22% for ADE and 11% for FDE, outperforming CAPHA, which reports the second-best performance. As for cyclists, we also report the second-best performance, achieving an ADE of 0.85 and an FDE of 2.03. Moreover, we report the detailed error values across various agent types and for all intersections within the InD dataset in Table IV. As demonstrated, in the Bendplatz scenario,

the model consistently shows high accuracy, when considering all agent types. It also showcases balanced performance across vehicle, pedestrian, and cyclist types, achieving an ADE of 0.23 and an FDE of 0.46. As for the other three scenarios, the results are notably better for vehicles and pedestrians than for cyclists, indicating the model proficiency in forecasting the motions of these agent types across different intersection scenarios. Furthermore, the results in Table IV suggest that predicting the motions of agents within the Strasse intersection is the most challenging

among all scenarios, as evidenced by higher ADE and FDE error values across various agent types. This suggests that our model faces increased complexity or unique dynamics within these intersections. It is worth mentioning that this challenge is not unique to our model, as it is consistently observed across other benchmark models, as shown in Table II.

Dynamic scenarios: Table V presents the performance comparison of our model and six benchmark models over the dynamic scenarios of the SDD dataset, all generating multimodal predictions. Our model stands out by outperforming all others in terms of ADE and off-road rate metrics and achieving the second-best result for the FDE error value. Notably, when compared to two transformer-based approaches, MID and MSN, our model demonstrates superior performance with an improvement of 15% in ADE and 13% in FDE. In contrast to P2T, which leverages an IRL-based approach, our model demonstrates substantial enhancements with a 26% improvement in ADE, a 25% improvement in FDE, and 49% improvement in off-road rate metrics. These enhancements highlight the effectiveness of integrating hierarchical transformers and occupancy maps in capturing complex scene contexts and producing more accurate, human-like motion predictions. Furthermore, in comparison to CAPHA, a similar approach employing both transformers and IRL-based techniques, HTMF showcases a 16% improvement in ADE, 8% in FDE, and a 16% reduction in off-road rate. When compared to another similar benchmark, Guo et al. [43], which also utilizes OGM predictions, HTMF achieves competitive FDE results while improving ADE by 4% and reducing the off-road rate by 53%.

Ablation study: We carry out an ablation analysis to explore the impact of major modules in our proposed motion forecasting model, namely the reward maps, policy maps, and transformer-based motion decoder. Accordingly, we train three alternative versions of our model, each without one or two of these crucial modules. First, to assess the impact of conditioning future motions on plans, we exclude the reward and policy map modules, opting instead for max-pooling and latent variables. Second, we eliminate the transformer-based motion decoder and replace it with a GRU-based motion decoder. Finally, we explore the model performance using behavior cloning, where feature maps are explicitly mapped to motions within each grid. Notably, all ablated models are evaluated using the same test settings over the SDD dataset. The results, reported in Table VI, reveal an overall performance reduction in the ablated versions across three metrics, including ADE, FDE, and off-road rate.

F. Qualitative Results

To intuitively assess the effectiveness of our introduced model, HTMF, we present visualizations of forecasting results from various scenarios in Fig. 3. For each scenario, we visualize agent-centric scene images, reward maps derived from the scene, final goals, paths to these goals, OGM prediction at $t = 8$, and multimodal predicted motions. Fig. 3 depicts the rewards maps, goal and path state visitation frequencies, and OGM predictions as heatmaps, where brighter colors represent higher values or occupancy probabilities. Notably, for visualization purposes,

we display $K = 5$ for the multimodality. The results highlight that HTMF excels in effective motion forecasting across diverse scenarios, producing varied multimodal outputs. Moreover, it demonstrates the effectiveness of the scene-compliant policy module, resulting in obstacle avoidance for the agents. Furthermore, the OGM predictions illustrate the future locations of agents as a probability heatmap at time $t = 8$. The OGM predictions presented in Fig. 3 represent the capability of our OGM decoder to accurately predict future position distributions that are consistent with the scene. The predicted OGMs also represent multimodality in several cases. This highlights HTMF capability in generating motions that conform to the environment, demonstrating its adaptability and competence in capturing complex interactions.

VI. CONCLUSION

In this work, we have developed a context-aware motion forecasting approach named HTMF, utilizing hierarchical transformers and MaxEnt IRL to generate scene-compliant motions for heterogeneous agents. HTMF formulates the motion forecasting task as an MDP, conditioning the future motions of agents on context information by finding the optimal policy and incorporating occupancy grid maps. Comprehensive experiments conducted on three large-scale datasets encompassing diverse scenarios, including highways and unsignalized intersections, demonstrate the higher performance of HTMF compared to state-of-the-art benchmarks. Qualitative analysis indicates that our model proficiently approximates the true distribution of potential motions, particularly for diverse agents such as cars and pedestrians. Ablation analysis further demonstrates that learning rewards from context information obtained from hierarchical transformers, the optimal policy, and transformer-based motion decoder modules of HTMF significantly enhance forecasting performance. The qualitative assessment also confirms our model ability to generalize effectively across unfamiliar contexts. It also indicates that the goal and path state visitation frequencies derived from human behavior are multimodal, illustrating the effectiveness of the MaxEnt IRL process. In future work, we aim to advance our motion forecasting model towards multi-agent scenarios by introducing a reference world frame for all agents. This extension will allow the model to query information from a shared frame, promoting a more collaborative forecasting approach. Furthermore, another future direction is to investigate the incorporation of vectorized representations for context information.

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